

OOPS ASSIGNMENT – 05

VIHAAN MITTAL – 1024150189

1.

```
#include <iostream>
```

```
using namespace std;
```

```
class Base
```

```
{
```

```
public:
```

```
    int x;
```

```
    void setX(int a)
```

```
    {
```

```
        x = a;
```

```
    }
```

```
};
```

```
class Derived : public Base
```

```
{
```

```
public:
```

```
    int y;
```

```
    void setY(int b)
```

```
    {
```

```
        y = b;
```

```

    }

    int sum()
    {
        return x + y;
    }
};

int main()
{
    Base b;
    b.setX(5);
    cout << b.x << endl;

    Derived d;
    d.setX(3);
    d.setY(4);
    cout << d.sum() << endl;

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
5
7
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

2.

```

#include <iostream>

using namespace std;

```

```
class Base
```

```
{
```

```
protected:
```

```
    int x;
```

```
};
```

```
class Derived : public Base
```

```
{
```

```
public:
```

```
    void set(int a)
```

```
{
```

```
        x = a;
```

```
}
```

```
    void show()
```

```
{
```

```
        cout << x << endl;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```
    Derived d;
```

```
    d.set(10);
```

```
    d.show();
```

```
    return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
10
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

3.

```
#include <iostream>
```

```
using namespace std;
```

```
class Base
```

```
{
```

```
public:
```

```
    int a = 1;
```

```
protected:
```

```
    int b = 2;
```

```
private:
```

```
    int c = 3;
```

```
};
```

```
class PublicDerived : public Base
```

```
{
```

```
public:
```

```
    void show()
```

```
    {
```

```
        cout << a << " " << b << endl;
```

```
    }
```

```
};
```

```
class ProtectedDerived : protected Base
```

```
{  
public:  
    void show()  
    {  
        cout << a << " " << b << endl;  
    }  
};
```

```
class PrivateDerived : private Base  
{  
public:  
    void show()  
    {  
        cout << a << " " << b << endl;  
    }  
};
```

```
int main()  
{  
    PublicDerived p;  
    p.show();  
  
    ProtectedDerived pr;  
    pr.show();  
  
    PrivateDerived pv;  
    pv.show();
```

return 0;

}

```
● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
1 2
1 2
1 2
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %
```

4.

#include <iostream>

using namespace std;

class A

{

public:

int x = 10;

};

class B : public A

{

};

class C : public A

{

};

class D : public B, public C

{

};

```
class E  
{  
public:  
    int y = 5;  
};
```

```
class F : public A, public E  
{  
};
```

```
class G : public B  
{  
};
```

```
int main()  
{  
    B b;  
    cout << b.x << endl;  
  
    F f;  
    cout << f.x + f.y << endl;  
  
    C c;  
    cout << c.x << endl;  
  
    G g;
```

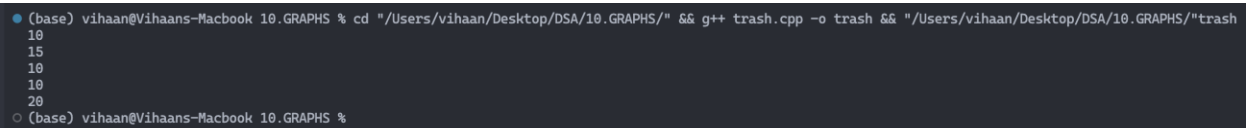
```
cout << g.x << endl;
```

```
D d;
```

```
cout << d.B::x + d.C::x << endl;
```

```
return 0;
```

```
}
```



```
(base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
10
15
10
10
20
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

5.

```
#include <iostream>
```

```
using namespace std;
```

```
class Base
```

```
{
```

```
public:
```

```
Base()
```

```
{
```

```
cout << "Base Constructor" << endl;
```

```
}
```

```
~Base()
```

```
{
```

```
cout << "Base Destructor" << endl;
```

```
}
```

```
};
```



```

class Derived : public Base
{
public:
    Derived()
    {
        cout << "Derived Constructor" << endl;
    }

    ~Derived()
    {
        cout << "Derived Destructor" << endl;
    }
};

```

```

int main()
{
    Derived d;

    return 0;
}

```

```

(base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
Base Constructor
Derived Constructor
Derived Destructor
Base Destructor
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

6.

```

#include <iostream>

using namespace std;

```

```
class Book
{
public:
    string title;
    string author;
    int price;

    void setBook(string t, string a, int p)
    {
        title = t;
        author = a;
        price = p;
    }

    void showBook()
    {
        cout << title << " " << author << " " << price << endl;
    }
};
```

```
class Textbook : public Book
{
public:
    string subject;

    void setTextbook(string t, string a, int p, string s)
    {
```

```

        title = t;
        author = a;
        price = p;
        subject = s;
    }

    void showTextbook()
    {
        cout << title << " " << author << " " << price << " " << subject << endl;
    }
};

int main()
{
    Book b;
    b.setBook("C++", "Bjarne", 500);
    b.showBook();

    Textbook t;
    t.setTextbook("Maths", "RD Sharma", 300, "Algebra");
    t.showTextbook();

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
C++ Bjarne 500
Maths RD Sharma 300 Algebra
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

7.

```
#include <iostream>  
using namespace std;
```

```
class Speedometer  
{  
public:  
    int speed;  
  
    void setSpeed(int s)  
    {  
        speed = s;  
    }  
};
```

```
class FuelGauge  
{  
public:  
    int fuel;  
  
    void setFuel(int f)  
    {  
        fuel = f;  
    }  
};
```

```
class Thermometer  
{
```

public:

int temp;

void setTemp(int t)

{

temp = t;

}

};

class CarDashboard : public Speedometer, public FuelGauge, public Thermometer

{

public:

void show()

{

cout << speed << " " << fuel << " " << temp << endl;

}

};

int main()

{

CarDashboard c;

c.setSpeed(80);

c.setFuel(50);

c.setTemp(30);

c.show();

```
    return 0;
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
80 50 30
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

8.

```
#include <iostream>
```

```
using namespace std;
```

```
class LibraryUser
```

```
{
```

```
public:
```

```
    string name;
```

```
    int id;
```

```
    string contact;
```

```
void setUser(string n, int i, string c)
```

```
{
```

```
    name = n;
```

```
    id = i;
```

```
    contact = c;
```

```
}
```

```
void showUser()
```

```
{
```

```
    cout << name << " " << id << " " << contact << endl;
```

```
}
```

```
};
```

```
class Student : public LibraryUser
{
public:
    int grade;

    void setStudent(string n, int i, string c, int g)
    {
        name = n;
        id = i;
        contact = c;
        grade = g;
    }

    void showStudent()
    {
        cout << name << " " << id << " " << contact << " " << grade << endl;
    }
};
```

```
class Teacher : public LibraryUser
{
public:
    string dept;

    void setTeacher(string n, int i, string c, string d)
    {
```

```

        name = n;

        id = i;

        contact = c;

        dept = d;
    }

    void showTeacher()
    {
        cout << name << " " << id << " " << contact << " " << dept << endl;
    }
};

int main()
{
    Student s;

    s.setStudent("Rahul", 1, "12345", 10);

    s.showStudent();


    Teacher t;

    t.setTeacher("Amit", 2, "67890", "CS");

    t.showTeacher();


    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
Rahul 1 12345 10
Amit 2 67890 CS
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %

```

9.


```
#include <iostream>
```

```
using namespace std;
```

```
class Vehicle
```

```
{
```

```
public:
```

```
    string make;
```

```
    string model;
```

```
    int year;
```

```
    void setVehicle(string m, string mo, int y)
```

```
{
```

```
        make = m;
```

```
        model = mo;
```

```
        year = y;
```

```
}
```

```
};
```

```
class Truck : public Vehicle
```

```
{
```

```
public:
```

```
    int load;
```

```
    void setTruck(string m, string mo, int y, int l)
```

```
{
```

```
        make = m;
```

```
        model = mo;
```

```
        year = y;
        load = l;
    }
};
```

```
class RefrigeratedTruck : public Truck
```

```
{
```

```
public:
```

```
    int temp;
```

```
    void setRefrigerated(string m, string mo, int y, int l, int t)
```

```
{
```

```
    make = m;
```

```
    model = mo;
```

```
    year = y;
```

```
    load = l;
```

```
    temp = t;
```

```
}
```

```
    void show()
```

```
{
```

```
        cout << make << " " << model << " " << year << " " << load << " " << temp << endl;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```

    RefrigeratedTruck r;

    r.setRefrigerated("Tata", "X1", 2022, 5000, -10);

    r.show();

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash
Tata X1 2022 5000 -10
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %

```

10.

```

#include <iostream>

using namespace std;

```

```

class Person

```

```

{
public:
    string name;
    string address;
};

```

```

class Staff : virtual public Person

```

```

{
public:
    int emp_id;
    string dept;
};

```

```

class Student : virtual public Person

```

```
{  
public:  
    int stu_id;  
    int grade;  
};  
  
class TeachingAssistant : public Staff, public Student  
{  
public:  
    void setData()  
    {  
        name = "Rahul";  
        address = "Delhi";  
        emp_id = 101;  
        dept = "CS";  
        stu_id = 202;  
        grade = 9;  
    }  
  
    void show()  
    {  
        cout << name << " " << address << " " << emp_id << " " << dept << " " << stu_id <<  
" " << grade << endl;  
    }  
};  
  
int main()
```

```
{  
  
    TeachingAssistant t;  
  
    t.setData();  
  
    t.show();  
  
  
    return 0;  
  
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash  
Rahul Delhi 101 CS 202 9  
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```